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AI-Supported Socratic Dialogue Agents vs. Generic AI Chatbots for Intermediate EFL Learners' Critical Thinking, Epistemic Beliefs, and Argumentation: Mixed-Methods Evidence

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ABSTRACT

This mixed-methods, quasi-experimental study investigated the impact of artificial intelligence (AI)-supported Socratic Dialogue Agents (SDAs) on critical thinking, epistemic beliefs, and argumentation skills among intermediate English as a Foreign Language (EFL) learners. Ninety male intermediate EFL learners from Iranian language institutes were randomly assigned to one of three conditions: (a) an AI-Socratic dialogue agent group, (b) a generic AI chatbot group, or (c) a no-AI control group. Quantitative data were collected through validated instruments before and after the intervention and then analyzed using mixed-design analyses of variance (ANOVAs). To complement these results, qualitative data from post-intervention focus groups (five participants per group) were examined through thematic analysis to elicit perceptions and through content-structural analysis to examine real-time English interactions. The quantitative findings indicated that the AI-Socratic group achieved notably greater gains in critical thinking, epistemic beliefs, and argumentation skills than the generic AI and control groups, while the qualitative findings helped to account for these outcomes by foregrounding greater cognitive challenge, an enhanced understanding of knowledge, the cultivation of advanced argumentation strategies, and sustained motivational engagement within the AI-Socratic condition; taken together, the findings carry theoretical implications in suggesting that AI-supported Socratic dialogue can embody principles of constructivist and epistemic scaffolding.

1 | Introduction

The Socratic method remains foundational to critical inquiry because its sequenced questions push learners to clarify meanings, justify positions, and revise ideas through reflection (Yang et al. 2005; Mahoney et al. 2023). In EFL settings, this kind of inquiry is typically enacted through argumentation, where learners formulate claims, support them with evidence, and evaluate alternatives, processes that are closely tied to epistemic beliefs about what counts as knowledge and how it should be warranted (Lee et al. 2016). As Mirzababaei and Pammer-Schindler (2021) illustrate, argumentation-oriented frameworks such as Bloom's taxonomy and Toulmin's (2003) model help to show how questioning can prompt evaluation and synthesis and make warrants more explicit. At the same time, epistemic beliefs shape how learners interpret uncertainty, appraise evidence, and respond to competing perspectives, which makes inquiry-based dialogue a plausible pathway for epistemic development (Huang et al. 2023; Knezic et al. 2010). However, sustaining Socratic dialogue is demanding, since it depends on skilled facilitation and

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repeated follow-up moves that are difficult to maintain in large classes (Ho 2024). This challenge is especially pronounced in large EFL classrooms, where the limited attention of the teacher may hinder the continuous inquiry required to facilitate evidence-based reasoning beyond mere linguistic accuracy (Lin and Hung 2025).

Due to these limitations, artificial intelligence (AI)-mediated instruction is increasingly regarded as a method to enhance interaction, feedback, and personalized prompting in EFL learning (Rezai et al. 2024; Yang and Yao 2025). As Díaz and Nussbaum (2024) note, advances in natural language processing (NLP) and generative conversational AI have enabled agents to sustain context-sensitive dialogue, while developments in neural models have strengthened the coherence and adaptivity of these exchanges (Zhai and Wibowo 2023). Studies also suggest that AI tutors can support engagement (Teng & Huang, 2026), metacognitive reflection (Teng & Shen, 2025), and motivation (Teng, 2025), when they provide responsive guidance across extended interaction (Yan et al. 2025). Still, the educational value of AI depends on interactional design rather than availability, since many chatbots prioritize information delivery and surface-level assistance over inquiry that requires justification and counter-consideration (Zhang et al. 2023). This concern has motivated growing interest in AI-supported Socratic Dialogue Agents (SDAs), which aim to emulate inquiry facilitation by evaluating learner contributions and generating targeted follow-up questions (Nahar 2025). Such agents may reduce dependency on continuous teacher orchestration and make inquiry-based dialogue more scalable across classrooms (Balbay 2019), while early work indicates that Socratic-questioning designs can promote reasoning and value negotiation in dialogue (Almatrafi et al. 2024) and support argumentative writing and critical thinking in language-learning settings (Chang et al. 2024). Lee and Wu (2025) similarly argue that structured questioning can be aligned with cognitive and epistemic goals, and recent higher-education evidence suggests that large language model (LLM)-based Socratic agents may foster reflective thinking compared with non-Socratic agents (Xi et al. 2025). Nevertheless, findings remain mixed for epistemic beliefs, and the mechanisms linking AI-mediated inquiry to higher-order outcomes are not yet well specified.

Accordingly, the present study addresses this gap through a three-group comparative design with Iranian intermediate EFL learners, comparing an AI-supported SDA condition, a generic AI chatbot condition, and a no-AI control condition. By combining quantitative change over time with qualitative evidence of learners' experiences, the study examines whether, and how, AI-supported Socratic inquiry contributes to critical thinking, epistemic beliefs, and argumentation skills under large-classroom constraints (Lee and Wu 2025; Lin and Hung 2025).

2 | Literature Review

2.1 | Key Constructs Involved in the Study

Socratic Dialogue has its roots in classical Greek philosophy, where Socrates relied on disciplined questioning to elicit definitions, expose contradictions, and advance understanding through inquiry rather than transmission (Redfield 2017). In Plato's dialogues, it is presented as a collaborative route toward warranted belief, in which interlocutors clarify terms, test assumptions, and refine reasoning as the exchange unfolds (Brickhouse and Smith 2004). Building on this tradition, educational accounts portray Socratic Dialogue as facilitator-guided talk that uses open-ended questions to prompt reflection and self-generated insight, distinguishing it from instruction that privileges recall over intellectual exploration (Davey Chesters 2012). Empirical work likewise suggests that structured questioning can strengthen analytical judgment and conceptual clarity across educational settings (Knezic et al. 2010), and Nussbaum et al. (2023) similarly connect dialogic questioning routines to learners' capacity to evaluate claims, weigh evidence, and articulate justifications.

Within this study, critical thinking, epistemic beliefs, and argumentation skills are treated as complementary outcomes that can be jointly cultivated through inquiry-based dialogue. Facione (1990) frames critical thinking as purposeful, self-regulated judgment involving interpretation, analysis, evaluation, inference, and explanation. Argumentation skills intersect with evaluative reasoning yet focus more directly on constructing and critiquing claims in discourse by coordinating evidence, warrants, and rebuttals, as captured in Toulmin's framework (Toulmin 2003). Epistemic beliefs shape learners' views of knowledge and knowing, spanning absolutist orientations grounded in authority to evaluativist positions that treat knowledge as provisional and evidence-based (Hofer and Pintrich 1997). With advances in language technologies, computational approaches to Socratic inquiry have moved from scripted, rule-based prompting (Noh et al. 2012) toward more flexible AI dialogue systems capable of sustaining exploratory exchanges and coherent questioning sequences (Yang et al. 2022). This shift is central here because it underscores that AI can support inquiry only when interaction design persistently presses learners to justify, qualify, and revise claims, rather than merely retrieve or summarize information.

2.2 | Theoretical Foundations of the Study

The study situates AI-supported Socratic dialogue within constructivist and dialogic accounts of learning. From a Vygotskian perspective, higher psychological functions develop through socially mediated interaction, and development is supported when assistance targets learners' emerging capabilities in the Zone of Proximal Development (ZPD) (Vygotsky 1978). Reznitskaya and Gregory (2013) further argue that dialogic pedagogy gives this view practical form by foregrounding talk that renders reasoning visible and open to negotiation. In a similar vein, Asterhan and Schwarz (2016) show that structured argumentative dialogue can operate as a scaffold for reasoning when interaction requires engagement with competing ideas and the defense of claims with grounds. Accordingly, Socratic Dialogue is framed here as an interactional scaffold that repeatedly makes thinking an explicit object of discussion, aligning with argumentation-for-learning perspectives.

Metacognitive and dual-process accounts help explain why Socratic questioning should influence reasoning quality. Metacognition concerns awareness and regulation of one's cognitive processes (Flavell, 1979), and learners benefit when prompts elicit monitoring and revision during complex tasks

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(Moritz & Lysaker, 2018). Socratic questioning supports this mechanism by directing learners to scrutinize the validity of inferences and the adequacy of evidence (Yang et al., 2005). Dual-process theory complements this view by distinguishing rapid intuitive responses from slower analytic evaluation, and prompts that require justification can encourage more reflective processing (Barrouillet, 2011), a point Darwin et al. (2024) also underscore for AI-mediated contexts. Epistemic cognition research further suggests that beliefs about knowledge shape how evidence is interpreted and uncertainty is managed (Bråten et al., 2022); Schiefer et al. (2021) describe meaningful differences in viewing knowledge as certain, subjective, or evaluative, and dialogic environments that demand justification can support more evaluative epistemic orientations (Ruwe & Mayweg-Paus, 2023). Toulmin's (2003) model clarifies the structure of warranted claims, while computational argumentation work shows how relations among claims, supports, and counterarguments can be systematically modeled (Craven & Toni, 2016).

In applied terms, AI-supported SDAs can instantiate these principles through structured questioning and argumentation routines aligned with Toulmin's components (Held et al., 2024) and through prompts adapted to learners' language level and content demands in EFL settings (Gregorcic et al., 2024), whereas generic chatbots often privilege conversational fluency or information provision over sustained epistemic probing (Alsaiari et al., 2025). Motivation is also relevant: Self-Determination Theory (SDT) links deeper engagement and persistence to autonomy, competence, and relatedness (Deci & Ryan, 2013), and technology-enhanced designs can support these needs through guidance and feedback (Chiu et al., 2024). Taken together, these frameworks justify why SDAs should be most effective when questioning remains cognitively demanding while also being motivationally supportive.

2.3 | Empirical Foundations of the Study

Despite strong theoretical arguments, the empirical base still leaves key questions unresolved in EFL settings. Comparative evidence that isolates Socratic questioning in AI from broader conversational AI use or human-led discussion remains limited, particularly for the combined outcomes examined here (Lin and Hung 2025; Yang et al. 2022). Although some studies report benefits of chatbots for EFL learning processes, the focus is often on language gains rather than higher-order cognitive outcomes, and control conditions are not always sufficient to attribute effects to inquiry-based mechanisms (Kohnke et al. 2023). Methodologically, the relative scarcity of mixed-methods designs that integrate standardized measures with discourse-level evidence constrains inferences about how gains are produced (Ouyang et al. 2022; Weinberger and Fischer 2006). Reviews also note risks associated with over-reliance on AI, including reduced independence and the uncritical acceptance of flawed outputs (Zhai and Wibowo 2023).

Contextual gaps remain as well. Research in non-Western EFL settings is still limited (Balbay 2019; Hwang et al. 2020), and language barriers can make conversations harder, which might affect how involved and successful learners are (Ho 2024). The affective dimension of inquiry also merits closer attention, since challenge can be productive or discouraging depending on calibration and proficiency (Michel-Villarreal et al. 2023). Ethical concerns about unverified information and its implications for learners' epistemic agency further make careful design and evaluation essential (Liu et al. 2024). Together, these gaps motivate a theoretically informed mixed-methods comparison to clarify whether AI-supported SDAs offer distinct advantages over generic chatbots and non-AI discussion for critical thinking, epistemic beliefs, and argumentation in EFL; accordingly, the study addresses the following research questions (RQs):

RQ1. To what extent does engaging with an AI-supported SDAs, in contrast to a standard AI chatbot and a control group without AI, influence the critical thinking abilities of EFL learners?

RQ2. To what extent does interaction with AI-supported SDAs, compared to other instructional methods, have a significant impact on the epistemic beliefs of EFL learners?

RQ3. In what ways does an AI-supported SDA enhance EFL learners' argumentation skills when compared to a generic AI chatbot and discussions without AI?

RQ4. How do AI-supported SDAs versus generic AI chatbots affect critical thinking, epistemic beliefs, and argumentation skills in EFL learners?

3 | Method of the Study

3.1 | Study's Design

This research employed a mixed-methods, quasi-experimental approach with three parallel instructional conditions, selected to examine in a comprehensive way how AI-supported SDAs shape intermediate EFL learners' critical thinking, epistemic beliefs, and argumentation skills. The quantitative component employed a pretest-posttest design to statistically assess group outcomes, whereas the qualitative component, utilizing focus groups and discourse analysis, offered a comprehensive depiction of learners' experiences and cognitive processes (Creswell and Poth 2016). Integrating the two evidence streams through methodological triangulation strengthened the validity and interpretive depth of the findings and supported a more textured understanding of the complex phenomena under investigation.

3.2 | Participants

The study involved 90 male intermediate EFL learners from Iranian language institutes, reflecting the national policy of gender-segregated classrooms. Participants were 16 to 29 years old (M = 22.4, standard deviation [SD] = 3.1), with an average of 4.2 years of prior English study, and their intermediate proficiency was verified via institutional records and the Oxford Quick Placement Test (scores 50–70). Recruited through convenience sampling from intact classes, learners were then randomly assigned to an AI-supported Socratic dialogue group (n = 30), a generic AI chatbot group (n = 30), or a no-AI control group engaged in peer discussion (n = 30) to strengthen internal validity. For the qualitative strand, five highly engaged participants from each condition (n = 15) were purposively selected for post-

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intervention focus groups to elicit richer experiential accounts. Ethical safeguards included written informed consent, voluntary participation, anonymity via pseudonyms, and encrypted data handling.

3.3 | Instruments

This research employed established quantitative and qualitative tools to collect the necessary data, as detailed below. The Watson-Glaser Critical Thinking Appraisal (WGCTA; Watson and Glaser 1980) was selected because it targets five aspects of critical thinking that are directly relevant to instruction: inference, recognition of assumptions, deduction, interpretation, and evaluation of arguments. It is worth noting that, to support self-report use in an Iranian EFL context, the original multiple-choice format was adapted to a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree) and administered in Persian to secure comprehension. A pilot study with 30 intermediate EFL learners yielded satisfactory internal consistency (Cronbach's α = 0.852), and three applied linguistics experts reviewed the items for relevance, recommending minor culturally oriented wording adjustments. Reliability remained robust in the main study (α = 0.865), in line with the instrument's established psychometric profile in educational research (Watson and Glaser 1980).

The Epistemic Beliefs Inventory (EBI; Schraw et al. 2012) was used to assess epistemic beliefs, namely how learners construe the nature of knowledge, its certainty, and the bases for justification, which aligns with the study's focus on how AI-mediated dialogue may shape cognitive epistemologies. Following earlier adaptations, the pool was reduced from 32 to 25 items spanning five dimensions: simple knowledge, certain knowledge, quick learning, omniscient authority, and innate ability, and responses were recorded on a 5-point Likert scale. A Persian version was piloted with 30 EFL learners and demonstrated acceptable internal consistency (α = 0.84); three educational psychology experts then reviewed the items for construct fidelity and clarity, leading to minor culturally oriented wording revisions. Reliability in the main study was strong (α = 0.87), and prior validation work reports domain-level coefficients ranging from α = 0.62 to 0.81.

The revised 13-item Argumentation Skills Test (AST; Kuhn and Udell 2003) was employed to evaluate argumentation skills, concentrating on learners' capacity to formulate, substantiate, and assess arguments in alignment with Toulmin's (2003) components (e.g., claims, evidence, and counterarguments). The items were rated on a 5-point Likert scale, with total scores ranging from 13 to 65. Higher scores indicated better proficiency. A Persian version was tested with 30 similar participants and found to have acceptable internal consistency (α = 0.81). Three experts in argumentation then reviewed the items in light of the underlying theoretical frameworks. They confirmed that the items aligned with the theoretical framework and suggested minor wording changes. Reliability remained satisfactory in the main study (α = 0.83), and the original AST is grounded in accounts of argument development that support its use in EFL contexts (Kuhn and Udell 2003).

Focus group interviews were used to elicit richer accounts of how each condition shaped participants' experiences of critical thinking, epistemic reflection, and argumentation skills, extending what could be inferred from quantitative outcomes. Five highly active learners from each group (n = 15) were purposively selected for post-intervention sessions to explore perceptions in depth, while recognizing the possibility of peer influence as a source of bias (Krueger 2014). Semi-structured prompts aligned with the research questions invited participants to recall moments when AI or discussion led them to reconsider beliefs, to reflect on whether their views of knowledge felt fixed or dynamic, and to describe how questioning affected their ability to construct evidence-based arguments and rebuttals. Interviews were conducted via Zoom (60–75 min) and permitted English–Persian code-switching; protocols included brief icebreakers, thematically focused discussion, and debriefing. Recordings were transcribed verbatim, including verbal and non-verbal elements, to support thematic analysis and strengthen triangulation across data sources.

3.4 | Interventions

Over an 8-week period, all groups worked through four instructional units (16 lessons) from Connect Book 3, ensuring a common curricular base. Classes met twice weekly for 90 min and followed a stable sequence: brief review, presentation of new content, guided practice, and discussion tasks. While objectives and materials (themes, vocabulary, grammar, and reading/listening passages) were held constant, the practice and discussion phases were implemented differently across conditions, allowing a focused comparison of dialogic approaches to higher-order outcomes while preserving core EFL content (Yang et al. 2022). To limit instructor effects, one experienced teacher delivered all sessions under a blinding procedure, with standardized directions for the non-discussion phases and condition-specific prompting guidance for discussion, so that observed differences could be more plausibly attributed to the instructional conditions rather than teacher variation (Creswell and Poth 2016).

During the 20 to 30 min discussion phase, Group A interacted with an AI-supported Socratic Dialogue Agent (SDA) built on an educational large language model (LLM). The agent pursued inquiry through adaptive, multi-turn questioning that responded to learner contributions in real time and repeatedly pressed for clarification, justification, and reconsideration, using prompts such as "What brings you to this conclusion?" and "Can you consider an alternative explanation?," and "How does your example support your argument?" Learners were expected to articulate reasons, entertain opposing positions, and make their thinking explicit; for instance, an initial opinion in an environmental reading could be taken up through requests for textual evidence, counterexamples, and implications, encouraging Toulmin-style reasoning (claims, data, and warrants). It is worth noting that the agent maintained interactional continuity while keeping linguistic demands at a moderate level so that probing did not overwhelm learners, a design choice consistent with emerging LLM-Socratic work (Gregorcic et al. 2024; Held et al. 2024). In short, the condition was designed to isolate the contribution of Socratic questioning within AI mediation and to distinguish its effects from non-Socratic approaches.

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In Group B (n = 30), participants interacted with a generic AI chatbot running on a comparable large language model but without Socratic scripting. The interaction relied on direct, content-focused prompts and brief follow-up questions that prioritized recall and basic comprehension over analytic probing, for example, "What is the main idea of the text?" and "Can you provide two examples from the lesson?" After grammar work, prompts typically shifted to sentence rephrasing or identifying key phrases, aligning with lower-order levels in Bloom's taxonomy and keeping task demands manageable. Sessions lasted 20–30 min and emphasized confirmation of understanding through short affirmations, with limited pressure for justification, evaluation, or counter-consideration. This pattern reflects common EFL chatbot use for routine practice and foundational skills (Kohnke et al. 2023; Zhai and Wibowo 2023), and it served as a contrastive AI condition to examine how prompting style within AI-mediated dialogue may shape outcomes (Yang et al. 2022).

In Group C (n = 30), the no-AI control condition consisted of instructor-facilitated, peer-led discussions directed by open-ended, non-AI prompts such as "What do you think about this topic?" and "Can someone add to what was said?" The instructor managed turn-taking, encouraged quieter participants, and redirected discussion when needed, particularly in culturally oriented listening tasks, while peers responded to one another through agreement, elaboration, or mild disagreement rather than systematic Socratic probing. Discussions lasted 20–30 min and were recorded to document facilitation patterns, with encouragement paired with minimal instructor input to support autonomy. This condition was intended to benchmark the role of AI while keeping the social basis of classroom interaction visible, even if its support is less uniform than AI mediation (Balbay 2019; Knezic et al. 2010). All sessions were conducted online, and the review and presentation phases were held constant across groups so that differences were confined to the discussion phase and its mode of mediation (Knezic et al. 2010).

3.5 | Data Analysis

Quantitative data were analyzed in the Statistical Package for the Social Sciences (SPSS; Version 26). Preliminary checks supported parametric testing: Shapiro–Wilk tests indicated acceptable normality across the dependent variables, and internal consistency was satisfactory for all instruments (Cronbach's $\alpha > 0.70$). Descriptive statistics (means, standard deviations) were computed for pretest and posttest scores, and a one-way analysis of variance (ANOVA) on the pretest scores confirmed baseline comparability across the three groups. The main analyses then used mixed-design ANOVAs (Group × Time) for critical thinking, epistemic beliefs, and argumentation skills, with Bonferroni-adjusted pairwise comparisons used to follow up notable interaction effects. Practical significance was indexed via partial eta squared (ηp^2).

The focus group interviews were analyzed through thematic analysis, following the six-phase framework outlined by Braun and Clarke (2006). The process began with familiarization, during which the researchers repeatedly read the interview transcripts, documented initial impressions, and noted early analytic observations. Next, initial codes were generated across the full dataset to capture salient features of participants' accounts, after which these codes were collated into candidate themes by assembling relevant extracts under each provisional theme. The analysis then moved into a reviewing phase, where themes were refined by checking coherence within the coded extracts and consistency with the dataset as a whole; this stage involved iterative discussion among the research team to ensure that the developing thematic map remained conceptually clear and aligned with the aims of the study. The subsequent phase focused on defining and naming themes, with attention to specifying each theme's scope, internal logic, and contribution to the emerging explanatory narrative in relation to the research questions. Finally, reporting involved selecting vivid and representative excerpts and integrating the qualitative interpretation with the quantitative findings to clarify how learners experienced the intervention and how these experiences related to observed changes. To support reliability, two researchers independently coded a segment of the data and achieved a high level of agreement (Cohen's $\kappa = 0.83$), with disagreements resolved through discussion. Trustworthiness was further strengthened through member checking, as a summary of the themes was shared with a small group of participants to confirm interpretive accuracy.

4 | Results

The findings are presented in the order of the study aims. Preliminary analyses first examined whether the three groups were comparable before the intervention, followed by outcome-specific analyses for critical thinking, epistemic beliefs, and argumentation skills. The qualitative findings are then used to clarify how the two AI-mediated conditions differed in learners' reported experiences. Before the main analyses, the dataset was screened to check that parametric assumptions were tenable. Normality for all dependent variables was supported through visual inspection (histograms and Q-Q plots) and confirmed via Kolmogorov–Smirnov and Shapiro–Wilk tests, with skewness and kurtosis falling within ±1, which is commonly taken to justify parametric procedures. Baseline comparability was then examined using one-way ANOVAs on pretest scores, a step that matters in quasi-experimental work because it strengthens the basis for causal attribution by reducing the likelihood that initial group differences, rather than the intervention, account for post-intervention outcomes.

Table 1 summarizes the one-way ANOVAs used to check baseline comparability across the three groups. For critical thinking (WGCTA pretest), the analysis was non-significant, $F(2, 87) = 1.36$, $p = 0.262$, with Group A scoring M = 206.27 (SD = 17.87), Group B M = 211.27 (SD = 21.05), and Group C M = 214.57 (SD = 19.85). A similar pattern was observed for epistemic beliefs (EBI pretest), $F(2, 87) = 0.18$, $p = 0.839$, with means of M = 70.20 (SD = 7.99), M = 68.83 (SD = 9.22), and M = 69.53 (SD = 9.46), respectively. For argumentation skills (AST pretest), the group effect was also non-significant, $F(2, 87) = 2.07$, $p = 0.132$, and the corresponding means were M = 32.43 (SD = 5.20), M = 34.37 (SD = 3.86), and M = 32.07 (SD = 4.95). Overall, the absence of pretest differences is consistent with effective random assignment and supports interpreting posttest variation in relation to the instructional conditions. This baseline equivalence also permits a clearer reading of the subsequent RQ-specific findings.

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TABLE 1 | Pretest descriptive statistics and ANOVA results for WGCTA, EBI, and AST.

Measure	Group	N	Mean	SD	SE	ANOVA F (2, 87)	Sig. (p)
WGCTA Pre	A	30	206.27	17.871	3.263	1.359	0.262
	B	30	211.27	21.050	3.843		
	C	30	214.57	19.854	3.625		
	Total	90	210.70	19.715	2.078		
EBI Pre	A	30	70.20	7.997	1.460	0.176	0.839
	B	30	68.83	9.222	1.684		
	C	30	69.53	9.457	1.727		
	Total	90	69.52	8.832	0.931		
AST Pre	A	30	32.43	5.197	0.949	2.071	0.132
	B	30	34.37	3.855	0.704		
	C	30	32.07	4.948	0.903		
	Total	90	32.96	4.759	0.502		

TABLE 2 | Pretest and posttest descriptive statistics for WGCTA, EBI, and AST.

Measure	Group	Pretest (M ± SD)	Posttest (M ± SD)
WGCTA	A	206.27 ± 17.87	250.27 ± 20.79
	B	211.27 ± 21.05	231.90 ± 21.38
	C	214.57 ± 19.85	224.83 ± 20.98
EBI	A	70.20 ± 7.99	89.97 ± 8.61
	B	68.83 ± 9.22	81.23 ± 10.31
	C	69.53 ± 9.46	74.60 ± 9.90
AST	A	32.43 ± 5.20	48.37 ± 5.32
	B	34.37 ± 3.86	42.83 ± 4.50
	C	32.07 ± 4.95	37.47 ± 5.02

Following the initial confirmation of baseline parity, the analysis turned to pretest and posttest change to characterize the intervention effects. The descriptive pattern suggests a clear ordering of gains, with the AI-Socratic condition (Group A) showing the most pronounced improvement. For RQ1, Group A showed the largest increase in critical thinking. For RQ2, Group A showed the strongest improvement in epistemic beliefs. For RQ3, Group A also showed the largest gain in argumentation skills. Thus, the descriptive results indicate that the AI-Socratic condition produced the clearest improvement across the three targeted outcomes.

Table 2 shows the same pattern across outcomes. For WGCTA (critical thinking), Group A increased from M = 206.27 (SD = 17.87) to M = 250.27 (SD = 20.79), Group B from M = 211.27 (SD = 21.05) to M = 231.90 (SD = 21.38), and Group C from M = 214.57 (SD = 19.85) to M = 224.83 (SD = 20.98). These scores indicate that RQ1 was supported descriptively, because the AI-Socratic group gained more in critical thinking than the generic AI and no-AI groups. For epistemic beliefs, Group A rose from M = 70.20 (SD = 7.99) to M = 89.97 (SD = 8.61), Group B from M = 68.83 (SD = 9.22) to M = 81.23 (SD = 10.31), and Group C from M = 69.53 (SD = 9.46) to M = 74.60 (SD = 9.90). These scores support RQ2 descriptively, because the AI-Socratic group showed the strongest positive shift in epistemic beliefs. For argumentation, Group A increased from M = 32.43 (SD = 5.20) to M = 48.37 (SD = 5.32), Group B from M = 34.37 (SD = 3.86) to M = 42.83 (SD = 4.50), and Group C from M = 32.07 (SD = 4.95) to M = 37.47 (SD = 5.02). These scores support RQ3 descriptively, because the AI-Socratic group showed the greatest improvement in argumentation skills. Overall, the descriptive gradient is most pronounced in the AI-Socratic condition.

To test whether these descriptive trends were statistically reliable, mixed-design ANOVAs were conducted (Time × Group) to examine main effects and interaction patterns. The main effects of Time indicated overall improvement, whereas the interaction terms captured condition-specific change trajectories. Accordingly, the Time × Group interactions provided the main statistical evidence for RQ1, RQ2, and RQ3.

As reported in Table 3, there was a significant main effect of Time for WGCTA, F (1, 87) = 1042.70, $p < 0.001$, $\eta p^2 = 0.923$; for EBI, F (1, 87) = 1048.39, $p < 0.001$, $\eta p^2 = 0.923$; and for AST, F (1, 87) = 1418.82, $p < 0.001$, $\eta p^2 = 0.942$. The main effect of Group was not significant for WGCTA, F (2, 87) = 1.52, $p = 0.225$, $\eta p^2 = 0.034$, but

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TABLE 3 | Mixed ANOVA results for WGCTA, EBI, and AST.

Measure	Effect	F(df)	P	η^2
WGCTA	Time	1042.70 (1, 87)	<0.001	0.923
	Group	1.52 (2, 87)	0.225	0.034
	Time × Group	166.48 (2, 87)	<0.001	0.793
EBI	Time	1048.39 (1, 87)	<0.001	0.923
	Group	5.95 (2, 87)	0.004	0.120
	Time × Group	122.56 (2, 87)	<0.001	0.738
AST	Time	1418.82 (1, 87)	<0.001	0.942
	Group	11.40 (2, 87)	<0.001	0.208
	Time × Group	140.68 (2, 87)	<0.001	0.764

it was significant for EBI, $F(2, 87) = 5.95$, $p = 0.004$, $\eta p^2 = 0.120$, and for AST, $F(2, 87) = 11.40$, $p < 0.001$, $\eta p^2 = 0.208$. Importantly, the Time × Group interaction was significant across all three outcomes, including WGCTA, $F(2, 87) = 166.48$, $p < 0.001$, $\eta p^2 = 0.793$, EBI, $F(2, 87) = 122.56$, $p < 0.001$, $\eta p^2 = 0.738$, and AST, $F(2, 87) = 140.68$, $p < 0.001$, $\eta p^2 = 0.764$. These interaction effects show that improvement differed by condition, with the strongest gains occurring in the AI-Socratic group across the three outcome measures.

Bonferroni-adjusted post hoc comparisons were then conducted to unpack the significant interaction effects. For RQ1, Group A's critical thinking gains were significantly larger than those of Group B ($p < 0.001$) and Group C ($p < 0.001$), while Group B showed a smaller advantage over Group C ($p = 0.042$). For RQ2, Group A showed significantly stronger improvement in epistemic beliefs than Group B ($p < 0.001$) and Group C ($p < 0.001$), with Group B also exceeding Group C ($p = 0.003$). For RQ3, Group A gained significantly more in argumentation skills than Group B ($p < 0.001$) and Group C ($p < 0.001$), and Group B also outperformed Group C ($p < 0.001$). These findings indicate that the AI-supported Socratic Dialogue Agent produced the strongest gains across all three quantitative outcomes.

One-way ANOVAs on the posttest means (Table 2) indicated persistent between-group differences for WGCTA, $F(2, 87) = 12.45$, $p < 0.001$; for EBI, $F(2, 87) = 18.72$, $p < 0.001$; and for AST, $F(2, 87) = 22.16$, $p < 0.001$. Planned contrasts showed that Group A outperformed both Groups B and C (all $p < 0.001$) and that Group B also exceeded Group C (all $p < 0.01$). This pattern is reflected in the posttest means, with WGCTA at 250.27 for Group A versus 231.90 for Group B and 224.83 for Group C, EBI at 89.97 versus 81.23 and 74.60, and AST at 48.37 versus 42.83 and 37.47. The posttest profile departs from the initial pretest equilibrium and is consistent with an advantage for the AI-Socratic condition. Taken together, the quantitative findings show that the AI-Socratic condition was most effective for critical thinking, epistemic beliefs, and argumentation skills.

5 | Qualitative Results

The qualitative analysis addressed RQ4 by explaining how learners experienced the AI-Socratic and generic AI conditions differently. It also helped clarify why the AI-Socratic group showed stronger quantitative gains. The results of the qualitative data analysis are structured around four main themes detailed below.

5.1 | Increased Cognitive Challenge and Depth in Critical Thinking

Participants in the AI-Socratic group frequently reported being encouraged to move past surface-level answers and engage in more profound analytical thinking. This theme clarifies the critical thinking findings by showing that the AI-Socratic agent repeatedly pushed learners to justify, compare, and reconsider their ideas. The cognitive challenges observed correspond with the group's notable quantitative improvements on the WGCTA, where the posttest mean was 250.27. A student shared:

> I used to just give a summary of the text, but the agent kept asking, 'Why does this assumption hold?' 'What if the proof goes against it?' It made me break down the reading into parts that I would not have done on my own.

One individual highlighted the flexible aspect of the challenge, remarking, "*The questions changed as I answered them, like a real debate, which made me think three steps ahead*." A third participant highlighted the cumulative effect, sharing, "*At first, it was annoying, but then I started to look forward to the questions. I was not just reading anymore; I was questioning the text myself*." In sharp contrast, those in the generic AI group characterized their experiences as requiring less cognitive effort, emphasizing tasks related to verification and recall. As one noted, "*It wanted to know, 'What is the main point?' So, I copied from the lesson. No pressure to think about why they matter*." This contrast explains why the AI-Socratic group showed stronger critical thinking development than the generic AI and no-AI groups.

5.2 | Heightened Epistemic Awareness and Shifts in Beliefs

The focus group data highlighted a significant change in the way AI-Socratic participants understood knowledge, offering a

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narrative explanation for their impressive performance on the EBI (Posttest M = 89.97). This theme clarifies the epistemic belief findings by showing that learners became more attentive to evidence, uncertainty, and justification. These learners expressed a shift from perceiving knowledge as a static collection of facts handed down by authority figures. A participant expressed this change: "*Is this textbook fact true in all situations, or only in certain ones?" the agent asked. I thought that knowledge was always changing, not fixed by experts. Now, I automatically check my sources*." This indicates a shift toward a more evaluative approach to knowledge. A fellow student illustrated this transformation with great clarity: "*I used to think that there was only one right answer for everything. I can now see a map of arguments, some of which are stronger than others, but they are always open to new evidence*." In contrast, individuals utilizing the generic AI chatbot did not experience any noticeable change, with one participant remarking, "*It confirmed what we learned in class. Knowledge felt like it was stuck, if the text says it, it is done*." This pattern explains why the AI-Socratic condition produced stronger gains in epistemic beliefs than the comparison conditions.

5.3 | Development of Argumentation Strategies and Structural Sophistication

The qualitative data revealed that the AI-Socratic agent played a significant role in helping students build more intricate arguments, which was reflected in Group A's higher performance on the AST (posttest M = 48.37). This theme clarifies the argumentation findings by showing that learners practiced linking claims, evidence, warrants, and counterarguments during dialogue. Participants shared their experiences of embracing the elements that constitute a strong argument. "*At first, I would say, 'social media hurts young people' without any proof*," one shared. "*The agent asked, 'What data?' 'How does it show harm?' 'Talk about the arguments against addiction.' By the middle of the study, my posts were naturally structured*." The reports from the generic AI group lacked a thoughtful integration of claims, data, warrants, and rebuttals. One participant described their experience as "list-like," explaining, "*It asked for 'two examples of benefits,' but there was no link to why or how*." One participant in the AI-Socratic discussion emphasized the importance of skill transfer: "*I even started to argue better with my friends. Now I automatically look for the weak point in my own argument*." This evidence explains why the AI-Socratic condition showed the strongest improvement in argumentation skills.

5.4 | Sustained Engagement as a Mediator of Effects

Throughout the focus-group interviews, individuals in the AI-Socratic group often expressed a strong sense of intrinsic motivation and engagement, which seems to play a crucial role in the improvements noted across all three quantitative areas. This theme clarifies RQ4 by showing that the AI-Socratic agent differed from the generic chatbot not only in content but also in interactional quality. The dialogue was described as "addictive" and "like a puzzle," which aided in maintaining the mental effort necessary for profound reasoning. One participant provided further insight: "*It did not seem like a job. I was really interested to see what question the AI would ask next and if I could answer it well. I looked forward to the challenge*." Another expressed, "*It felt like a game I wanted to win, not a lesson I had to finish, even when it was hard." That push made me keep going back and thinking more*." A participant from the generic AI group remarked on the transactional aspect of their engagement: "*It helped me check my answers, but it got boring after a while. There was no surprise and no real talk*." Overall, the qualitative findings show that the AI-Socratic condition supported learners through sustained questioning, evidence-oriented reflection, structured argument practice, and continued engagement, whereas the generic AI chatbot mainly supported checking, recall, and short responses.

6 | Discussion

The study provides strong evidence that AI-supported SDAs function as effective educational tools, outperforming both a generic AI chatbot and peer discussion in supporting critical thinking, more complex epistemic beliefs, and argumentation skills among intermediate EFL learners. In addition to recording outcomes, the findings indicate the mechanisms by which structured, inquiry-based AI interaction can facilitate cognitive and epistemic advancement. Across data sources, the pattern converges on a coherent account: the AI-Socratic condition was associated with sustained cognitive challenge, more deliberate knowledge building, and repeated practice in constructing and evaluating arguments, and these features coincide with gains in transferable higher-order skills.

The significant gains in critical thinking (RQ1) in the AI-Socratic condition can be interpreted in light of the agent's role as an automated yet flexible cognitive stimulator. Unlike a generic chatbot that typically supports information retrieval and confirmation, the Socratic agent sustained engagement through open-ended questioning that invited learners into the ZPD (Vygotsky 1978) and required them to articulate reasons, interrogate claims, and revisit initial interpretations. Qualitative accounts suggest that this task was experienced as active work, with learners describing being compelled to deconstruct readings and question the text, moving beyond summarizing toward analysis and evaluation. This pattern aligns with cognitive load perspectives in which the aim is not simply to minimize load but to direct cognitive resources toward germane processing, such as evaluating assumptions, drawing inferences, and constructing justifications (Sweller 2011). By calibrating prompts to learners' proficiency, the SDA also helped manage intrinsic demands, enabling participation in the kind of purposeful, self-regulated judgment central to critical thinking (Facione 1990). While structured questioning is sometimes assumed to constrain creativity, the present findings point in the opposite direction, suggesting that Socratic scaffolding provided a stable frame that supported more systematic and, in practice, more creative engagement with complex ideas, consistent with emerging evidence on AI-assisted argumentation (Lin and Hung 2025; Zhang et al. 2025).

The shift in epistemic beliefs (RQ2) observed in the AI-Socratic condition illustrates how AI-mediated dialogue can shape learners' underlying conceptions of knowledge. The SDA's ongoing demand for justification seems to have pushed learners toward a more evaluativist view, in which knowledge is seen as contingent,

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based on evidence, and open to criticism. Learners' comments about seeing "a map of arguments" rather than "one right answer" capture this reorientation and align with epistemic cognition accounts suggesting that development is prompted when experiences disrupt absolutist assumptions and foreground evaluation (Hofer and Pintrich 1997). In this sense, the Socratic agent repeatedly modeled an analytic, curiosity-driven stance by pushing learners to warrant claims and consider alternatives. Conversely, the generic chatbot's propensity to validate authoritative "facts" may have fostered a more reductive and absolutist perspective on knowledge, highlighting the variability in epistemic quality of AI interactions, as suggested in previous studies (Huang et al. 2023; Schiefer et al. 2021). The implication is that epistemic growth is less a function of AI availability than of pedagogical design, with Socratic dialogue functioning as a practical route toward epistemic maturity (Ruwe and Mayweg-Paus 2023).

The AI-Socratic agent's Toulmin-informed design (Toulmin 2003) appeared to exert its strongest influence on argumentation skills (RQ3). Qualitative accounts suggest a clear process of internalization: learners moved from unsupported assertions toward arguments that more routinely integrated evidence, warrants, and counterarguments. In practical terms, the agent made argumentation theory usable by translating abstract components into concrete moves that could be enacted during discussion. This kind of structured support is particularly consequential in EFL contexts, where articulating complex reasoning through a second language can constrain what learners are able to express. Notably, participants described carrying these routines into everyday exchanges (e.g., "I even started arguing better with my friends"), pointing to a meaningful and functional learning outcome. The pattern aligns with work emphasizing the value of explicit scaffolding for developing argument literacy (Weinberger and Fischer 2006) and also helps explain why peer discussion, while socially engaging, yielded less advanced argumentation when sustained, structured focus was less consistent.

With respect to RQ4, the qualitative accounts of sustained engagement help explain the quantitative gains by pointing to motivation as a plausible mediator. Learners described the AI-Socratic dialogue as "addictive" and "like a puzzle," language that suggests strong intrinsic involvement. This pattern aligns with SDT (Deci and Ryan 2013): the agent can encourage independence by letting learners think for themselves in a guided conversation, help them improve through increasingly difficult questions that show their progress, and create a sense of connection through a supportive dialogue partner. In contrast to the more transactional feel of typical AI interactions, the AI-Socratic condition appears to have combined demanding questioning with the experience of psychological support, and this combination may have sustained the cognitive engagement required for deeper learning.

7 | Conclusions and Implications

This study examined how an AI-supported SDA shapes intermediate EFL learners' critical thinking, epistemic beliefs, and argumentation skills. Across outcomes, the AI-Socratic group showed the largest gains, and the qualitative evidence helps to clarify what these gains looked like in practice, including more deliberate reasoning, tighter integration of evidence, and more frequent attention to counterarguments than was evident in the generic AI and no-AI conditions. The findings highlight the SDA's contribution to higher-order development through adaptive inquiry.

Theoretically, this work extends constructivist perspectives by positioning AI as a flexible support system that can amplify learners' movement within the ZPD in dialogic EFL learning (Vygotsky 1978). By showing rapid shifts from more absolutist to more evaluativist epistemic beliefs under sustained AI probing, the study adds to epistemic cognition accounts and addresses a persistent gap in the technology literature regarding epistemic effects (Hofer and Pintrich 1997). It also situates AI-supported dialogue within argumentation theory by illustrating how warrants and rebuttals can be prompted and coordinated in interaction, linking computational argumentation with processes of digital knowledge construction (Craven and Toni 2016; Toulmin 2003; Weinberger and Fischer 2006). Grounded in a mixed-methods design, the study further speaks to metacognitive theory and SDT by highlighting how inquiry-oriented interaction can support monitoring and revision while sustaining autonomy, competence, and relatedness in learning (Deci and Ryan 2013; Flavell 1979; Moritz and Lysaker 2018). In this way, the findings align with emerging discussions of pedagogical intelligence in AI and point to a contemporary instantiation of Socratic methods in EFL (Díaz and Nussbaum 2024).

For EFL teachers, the results indicate that having weekly AI-Socratic sessions can be a helpful addition, along with training on how to adjust prompts to different skill levels and use available data to track student progress. At the classroom level, collaborative teacher networks may help normalize reflective routines that move beyond rote practice. For learning-material design, this indicates the need to develop flexible, multilingual resources that combine language learning with exploring different cultures, while also including checks for bias and clear information to ensure ethical and culturally aware use. At the policy level, focused funding for AI-in-EFL pilot projects, along with clear evaluation guidelines, would help ensure responsible use and show how skills are transferred.

8 | Limitations and Directions for Further Studies

Despite the study's use of a mixed-methods quasi-experimental design with methodological triangulation, it is important to acknowledge several limitations. The exclusively male sample, shaped by Iran's gender-segregated classroom policies, constrains generalizability to mixed-gender or culturally diverse contexts where interaction dynamics may differ. The eight-week timeframe constrains conclusions about long-term retention, as it may attenuate short-term gains without ongoing support. In addition, the quasi-experimental organization of participants within intact classes, while ethically and practically appropriate, may introduce group-level confounders that remain a concern even after baseline equivalence checks, and this necessarily tempers causal inference. Future research may tackle these concerns via cross-cultural comparative methodologies that investigate AI agents in coeducational settings, incorporating virtual reality simulations of inclusive dialogue to analyze the influence of gender on

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epistemic engagement. Longitudinal designs could track the everyday application of skills through wearable AI devices or professional communication applications, potentially using gamified incentives to sustain participation, while larger-scale trials might strengthen causal attribution via individual-level randomization within adaptive platforms and more secure cross-study data sharing through blockchain-supported infrastructures.

Declaration of AI Use

To enhance the manuscript's clarity and structure, we utilized Grammarly and GPT-5.5 for language editing and improving readability. The authors conducted all aspects of the study design, data collection, statistical analysis, and theoretical contributions, and they fully assume responsibility for the final content presented.

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